MIL-C-43892A
21 December 1984
SUPERSEDING
MIL-C-43892
1 May 1974

MILITARY SPECIFICATION

CLOTH, TWILL: COTTON AND NYLON, WATER REPELLENT TREATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of cotton and nylon blend twill cloth.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-1134 - Packaging of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

MILITARY

MIL-T-43548 - Thread, Polyester, Cotton-Covered and Rayon-Covered

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Laws and Regulations

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Copies may be obtained from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

3.1 Standard sample. The dyed and finished cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Cotton. The cotton shall be carded and drawn.

3.2.2 Nylon. The nylon shall be first quality, high tenacity, semi-dull staple having a nominal cut length of 1-1/2 inches and a round cross-section with a nominal denier of 2.25 to 2.50.

3.2.3 Yarns. The yarns shall be a blend of 50 percent cotton and 50 percent nylon drawn and spun into singles yarn for both the warp and filling, when tested as specified in 4.2.3. Percentages are based on the dry weight of the desized specimen. Tolerances will be limited to plus 5.0 percent nylon and minus 5.0 percent cotton.

3.3 Color. The color of the dyed and finished cloth shall be Olive Green 107 and shall match the standard sample (see 6.3 and 6.5). The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.2.3. When a standard sample is not available, the dyed cloth shall show no more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.2.3.

* 3.3.1 Matching. The color of the dyed and finished cloth shall match the standard sample under artificial daylight having a correlated color temperature of 7000 ± 500 K and shall be a good approximation to the standard sample under incandescent lamplight at 2850 ± 100 K.

* 3.3.2 Colorfastness. The dyed and finished cloth shall show fastness to laundering (after three cycles), light and crocking equal to or better than the standard sample when tested as specified in 4.2.3. When a standard sample is not available, the dyed and finished cloth shall show "good" fastness to laundering (after 3 cycles), perspiration, and light after 80 Standard Fading Hours; and shall show an AATCC Chrometic Transference Scale rating for crocking not lower than 3.5 when tested as specified in 4.2.3.

3.4 Physical requirements. The physical requirements of the dyed and finished cloth shall be as specified in table I when tested as specified in 4.2.3.

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TABLE I. Physical requirements

Weight oz./sq.yd. min.	Weight oz./sq.yd. max.	Yarns per inch minimum Warp	Yarns per inch minimum Filling	Breaking strength lbs., minimum Warp	Breaking strength lbs., minimum Filling	Air permeability (max.) cu. ft./min./ft^2
4.8	5.3	94	46	155	65	75.0

3.4.1 Weave. The weave shall be a $\frac{2}{1}$ right hand twill.

* 3.4.2 Width. The width of the cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages when fly shuttle looms are used, and exclusive of selvages and fringes when shuttleless looms are used.

3.5 Finish. The cloth shall be mercerized, dyed, and closely singed.

3.5.1 Nonfibrous material. Prior to the application of the water-repellent treatment, the starch and protein content, including chloroform-soluble and water-soluble material of the cloth shall not exceed 2.0 percent when tested as specified in 4.2.3.

3.6 pH. The pH value of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.2.3.

* 3.7 Water-repellent treatment. The cloth shall be given an approved water-repellent treatment (see 6.6) and shall conform to the requirements of 3.7.1. The water-repellent shall consist of aluminum salts of saturated carboxylic acids (such as formate, acetate, palmitate or stearate) mixed with refined mineral vegetable waxes. The product shall be applied either in the form of an aqueous emulsion or in the form of water-free solvent solution.

3.7.1 Spray rating. The results of the three individual determinations on the sample unit for spray rating shall be equal to or better than 90, 90, 80 when tested as specified in 4.2.3.

3.8 Dimensional stability. The cloth shall not shrink nor elongate more than 2.0 percent in either the warp or the filling direction when tested as specified in 4.2.3. The preshrinking process used shall not be identified by name or trademark, either on the cloth, ticket or package.

3.9 Seam efficiency. The finished cloth shall have a seam efficiency of not less than 80 percent when tested as specified in 4.2.3.

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* 3.10 Length and put-up. Unless otherwise specified (see 6.2), the finished cloth shall be furnished in continuous lengths each not less than 40 yards. The cloth shall be put-up in full width rolls as specified in 5.1.

* 3.11 Fiber identification. Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

* 3.12 Workmanship. The finished cloth shall conform to the quality of product established by this document. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the applicable established maximum point values.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.2.2 End item examination.

4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only (twill weave going to the right). When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in Section I of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1, except that only those slubs and

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knots which exceed the limits shown on Sears Fabric Defect Scales (see 6.7), "F" for slubs and "E" for knots, shall be scored. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 30 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 45 points. If one roll exceeds 45 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality. The lot shall be unacceptable if one or more rolls in the second sample exceeds 45 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to and including 3 inches in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth
Width less than specified
Uneven weaving
Poor dye penetration, mottled, streaky or cloudy
Not closely singed

4.2.2.2 Length examination.

4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

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4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

* 4.2.2.3 Shade and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade and appearance. If any roll in the sample is off shade; shaded side to side, side to center, or end to end; or does not have the same appearance as the standard sample, the lot represented by the sample shall be rejected.

* 4.2.2.4 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined for proper identification. The lot shall be unacceptable if two or more rolls in the sample are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.2.3 End item testing. The cloth shall be tested for the characteristics listed in table II. Methods of testing specified in FED-STD-191 wherever applicable and as listed in table II shall be followed. The physical and chemical values specified in Section 3 apply to the results of the determinations made on the sample unit for test purposes as specified in the applicable test method. The sample unit shall be 5 continuous yards full width of the finished cloth and 1/4 yard of the cloth prior to the application of the water-repellent treatment. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (number of sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE II. End item tests

Characteristic	Requirement paragraph	Test method
Cotton identification	3.2.1	1/
Nylon:		
Identification	3.2.2	1/
Luster	3.2.2	1/
Denier	3.2.2	1/
Fiber content:		
Nylon percent	3.2.3	2530
Cotton percent	3.2.3	2530 2/
Presence of labile sulfur	3.3.	2020
Colorfastness to:		
Laundering (after 3 cycles)	3.3.2	5610 3/
Perspiration	3.3.2	5680
Light	3.3.2	5660
Crocking	3.3.2	5651
Weight	3.4	5041
Yarns per inch	3.4	5050
Breaking strength	3.4	5100
Air permeability	3.4	5450
Weave	3.4.1	Visual 4/
Finish:		
Mercerized	3.5	1/
Singed	3.5	1/
Nonfibrous material	3.5.1	2611
pH	3.6	2811
Water repellency, spray rating	3.7.1	5526
Dimensional stability	3.8	5550
Seam efficiency	3.9	5110 5/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ The cotton (percent) shall be calculated as follows:
Percent cotton = 100 - percent nylon

3/ The specimens must be dried after each of the 3 laundering cycles.

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4/ One determination per sample unit and the result reported as "pass" or "fail".

5/ The needle shall measure 0.040 ± 0.001 inch across the blade at the eye. The thread shall be polyester/cotton or rayon-covered in accordance with MIL-T-43548, ticket No. 50, 2 or 3 ply for the needle and ticket No. 70, 2 or 3 ply for the looper.

* 4.2.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1134.

* 5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. The cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Levels A, B or Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth is intended for use in clothing items.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Width of cloth required (see 3.4.2).
c. Length of roll required if other than specified (see 3.10).
d. Selection of applicable levels of put-up and preservation and packing (see 5.1 and 5.2).

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6.3 Standard sample. For access to standard sample, address the contracting activity issuing the invitation for bids.

6.4 Cloth description. The cloth described in this document was woven using 22/1 yarn for both the warp and filling.

6.5 Suggested dyestuff formulations. A suggested but not mandatory dyestuff formulation for Olive Green 107 is as follows:

Cotton	Vat Green 3, CI 69500 Vat Black 25, CI 69525 Vat Green 8, CI 71050 Neutral Premetallized Olive Green BA
Nylon	Acid Brown 223 Neutralized Premetallized Yellow NW

Another formulation which has yielded satisfactory dyeing and which has the control advantages of less cross-staining is as follows:

Cotton	Vat Orange 15, CI 69025 Vat Green 6G Vat Black 25, CI 69525
Nylon	Acid Green 58 Acid Orange 64 Acid Blue 165 Acid Orange 60

*6.6 Water repellent treatment. Approval of the water repellent treatment is the responsibility of the US Army Natick Research and Development Center, Natick, MA 01760-5014, and is based on extensive tests including those for toxicity, which are not set forth in this document. Because of the time necessary to conduct full evaluation, only those chemical treatments already approved and so listed in the invitation for bid or request for proposal shall be considered acceptable for the related procurement.

6.7 Fabric defect scales. Fabric Defect Replica Kits are available from:

Sears Roebuck and Company
Department 817, (ATTN: BSC 23-29)
Sears Tower
Chicago, IL 60684

6.8 Changes from previous issue.. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review activities:

Army - MD
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DLA - CT

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-43892A	Cloth, Twill: Cotton and Nylon, Water Repellent Treated

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*. _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 20 Nov 84

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